

# ECON 0100 | Vignette B5 | Market Changes

## I Solutions

*Answers and work are shown in the red blocks.*

Members of the wizarding world have preferences for pumpkin pasties according to the following demand curve and production costs according to the following supply curve.

$$P = 14 - \frac{1}{2}Q_D \quad \text{and} \quad P = 2 + \frac{1}{2}Q_S$$

Prices are in galleons and quantity is in pasties.

From the curves,  $Q_d = 28 - 2P$  and  $Q_s = 2P - 4$ . Equilibrium is  $Q^* = 12$  pasties at  $P^* = 8$  galleons.

### Q1 | Price Elasticity

a) When price changes from 6 to 8 galleons:

What is the price elasticity of demand? \_\_\_\_\_

Is demand elastic, unit elastic, or inelastic between these prices? \_\_\_\_\_

-1, unit elastic. At 6 galleons  $Q_d = 16$ ; at 8,  $Q_d = 12$ .

$$\varepsilon = \frac{(12 - 16)/14}{(8 - 6)/7} = \frac{-2/7}{2/7} = -1$$

b) When price changes from 4 to 6 galleons:

What is the price elasticity of supply? \_\_\_\_\_

Is supply elastic, unit elastic, or inelastic between these prices? \_\_\_\_\_

$\frac{5}{3}$ , elastic. At 4 galleons  $Q_s = 4$ ; at 6,  $Q_s = 8$ .

$$\varepsilon = \frac{(8 - 4)/6}{(6 - 4)/5} = \frac{2/3}{2/5} = \frac{5}{3}$$

## Q2 | Supply & Demand Shifters

For each change below, which curve in the pumpkin pasty market shifts, and in which direction?

a) Recent drier growing seasons have made growing pumpkins more difficult.

*demand shifts up · demand shifts down · supply shifts up · supply shifts down*

■ Supply shifts up: pumpkins are harder to grow, which raises the cost of every pasty.

b) A popular channel on the wizarding social network FlueTube has been promoting pumpkin pasties, leading to an increase in the popularity of the snack.

*demand shifts up · demand shifts down · supply shifts up · supply shifts down*

■ Demand shifts up: a change in tastes raises what buyers will pay for every pasty.

c) An early freeze damaged the lemon crop, driving the price of lemons up and increasing the cost of lemon cakes.

*demand shifts up · demand shifts down · supply shifts up · supply shifts down*

■ Demand shifts up: lemon cakes are a substitute, so pricier lemon cakes send buyers to pasties.

d) The price of butter beer, which wizards drink with their pasties, rises.

*demand shifts up · demand shifts down · supply shifts up · supply shifts down*

■ Demand shifts down: butter beer is a complement, so pricier butter beer means fewer pasties at every price.

e) A boom at Gringotts raises wizards' incomes, and pumpkin pasties are an inferior good.

*demand shifts up · demand shifts down · supply shifts up · supply shifts down*

■ Demand shifts down: buyers of an inferior good switch to something better when income rises.

### Q3 | Comparative Statics

Each part below has two changes at once. Use a graph to show both shifts, then say what happened to the pumpkin pastie market without using numbers.

a) With both the drier growing seasons and the FlueTube promotion, use the graph above to discuss how the market has been impacted.

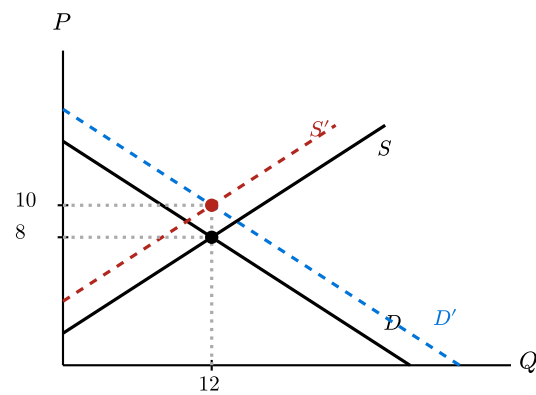
Prices: *increased* · *stayed the same* · *decreased* · *indeterminate*

Quantity: *increased* · *stayed the same* · *decreased* · *indeterminate*

Prices: increased. Quantity: indeterminate.

The drought shifts supply up, which raises the price and lowers the quantity. The promotion shifts demand up, which raises both. Both push the price up; they pull the quantity in opposite directions.

In the graph both shift up by 2: the price rises from 8 to 10 and the quantity stays at 12. A bigger promotion would raise the quantity, a worse drought would lower it.



b) A record pumpkin harvest makes pumpkins cheap for bakers. At the same time, Hogwarts adds a second Hogsmeade weekend each month, bringing more students to the village.

Prices: *increased* · *stayed the same* · *decreased* · *indeterminate*

Quantity: *increased* · *stayed the same* · *decreased* · *indeterminate*

Prices: indeterminate. Quantity: increased.

The cheap pumpkins shift supply down, which lowers the price and raises the quantity. More students shift demand up, which raises both. Both push the quantity up; they pull the price in opposite directions.

In the graph both shift by 2: the quantity rises from 12 to 16 and the price stays at 8.

